

Building on these exposure measures, we estimate a series of difference-in-differences models around the introduction of ChatGPT that exploit firm- and occupation-level variation in LLM exposure to estimate the causal effects of the ChatGPT shock on employment, wages, pay dispersion, job seniority, workplace arrangements, and overall hiring dynamics.

Our results show a clear impact of LLM introduction on UK labor markets, with effects emerging gradually across multiple dimensions of employment. At the firm level, a one standard deviation increase in LLM exposure is associated with a 0.3 percent reduction in total employment. The impact is concentrated among junior positions, with low-seniority employment declining by 0.4 percent while high-seniority positions show smaller and partly insignificant reductions. This compositional shift reduces the fraction of low-seniority workers, which mechanically raises average compensation in exposed companies as the workforce skews toward higher-paid senior employees.

These patterns also show up in hiring intentions. In an analysis of firm-level job posting data, we find that a one standard deviation increase in LLM exposure reduces the likelihood that firms post job vacancies in a given month by 1.0 percentage point. The reduction in job postings among exposed firms is strongest for technical and creative roles such as software engineers and designers. At the occupation level, we observe even more pronounced effects on hiring. A one standard deviation increase in an occupation’s LLM exposure is associated with a 3.9 percent decline in job postings, accompanied by substantial reductions in posted salaries.

Importantly, these effects are not uniformly distributed across the labor market. Our heterogeneity analysis shows that LLM disruption is concentrated almost entirely in higher-wage segments. High-compensation firms experience employment reductions of 0.7 percent per standard deviation increase in exposure, compared with near-zero effects for low-compensation firms. Similarly, high-salary occupations experience 5.7 percent declines in job postings per standard deviation, while low-salary roles show no significant change. These results point in the opposite direction from historical automation experiences, in which routine middle-skill occupations bore the brunt of technological displacement (Autor, 2024).

To interpret these results, we draw on the task-based framework (Autor et al., 2003; Acemoglu and Restrepo, 2018, 2019, 2020; Autor and Thompson, 2025). This framework models production as discrete tasks allocated between labor and capi-

2.1 Job Profile Data

The first dataset consists of all publicly available professional LinkedIn profiles between January 2021 through June 2025. The dataset includes approximately 75 million employment spells from 23 million unique individuals across 2.5 million UK-based firms. Each observation includes the worker’s job title, employer, start and end dates (if available), seniority level,³ and compensation.⁴ The data provide near-real-time employment stocks, which enable us to track within-firm workforce composition changes over time.

From the panel of individual LinkedIn profiles, we construct a monthly firm-level panel from January 2021 through June 2025. For each firm-month, we calculate total employment as the count of unique individuals with active employment spells at a certain company in that month, and log-transform this variable. This is our main employment stock measure. We then separate total employment into junior and senior employment stocks based on Revelio seniority classifications of jobs. We also calculate average compensation as the log of the mean of reported salaries plus bonuses in a firm. We only focus on firms that remained in business throughout the sample period.

To calculate firm-level exposure scores, we use the assessment framework developed by Eloundou et al. (2024). They evaluate 19,265 detailed work activities and tasks from the O*NET 27.2 database to determine whether LLM assistance could reduce task completion time by at least 50% while maintaining quality. We focus on their ‘E1’ measure, which captures exposure to current LLMs, rather than LLMs combined with additional software. We do so because it is the most appropriate metric for identifying short-term labor market effects that do not require additional software development or complex organizational changes. To obtain our firm-level

³Revelio Labs determines seniority using an ensemble model that combines three components: (i) current job characteristics including title, company, and industry; (ii) employment history including tenure and previous position seniority; and (iii) individual age. These three scores are averaged to create a continuous seniority metric, which is then mapped to seven ordinal levels based on keyword benchmarks (e.g., “junior,” “senior,” “director”). The levels range from Entry Level (e.g., interns, trainees) through Junior, Associate, Manager, and Director levels, to Executive and Senior Executive levels (e.g., C-suite positions).

⁴Revelio Labs estimates compensation using a predictive model trained on publicly available visa application data, self-reported salaries, and job posting information. The model incorporates position-specific variables (job title, seniority level, company, location), individual-specific factors (tenure at company), and temporal controls (year of observation) to generate salary predictions for positions where compensation data are not directly observed.

ket characteristics. We examine heterogeneity along two dimensions: compensation levels (splitting firms and occupations by median pre-treatment salary) and geographic concentration (comparing London-based versus non-London entities). These dimensions are highly policy-relevant, as compensation-level heterogeneity provides a first indication as to whether LLMs may compress or widen wage inequality, while geographic variation tests whether disruption concentrates in London’s knowledge economy or extends throughout the UK labor market.

4.3.1 Heterogeneity by Compensation Level

Table A2 examines how firm-level employment effects vary by average compensation, with high-compensation jobs serving as a proxy for more highly educated white collar work, and low-compensation jobs for relatively low-educated blue collar work. We split firms at the median pre-treatment compensation level. Panel A shows that high-compensation firms experience substantially larger employment reductions than their lower-paying counterparts. A one standard deviation increase in LLM exposure reduces total employment at high-compensation firms by 0.7 percent ($p = 0.009$), driven primarily by a 0.8 percent reduction in low-seniority positions ($p = 0.005$). In contrast, Panel B indicates that low-compensation firms show no significant employment effects, with point estimates near zero for both total employment and low-seniority positions.

This pattern extends to hiring intentions. Table A4 shows that a one standard deviation increase in LLM exposure reduces the probability of posting vacancies by 1.3 percentage points at high-compensation firms ($p < 0.001$), compared to a 0.7 percentage point reduction at low-compensation firms ($p = 0.001$). While both groups curtail hiring, the effect is approximately twice as large at high-compensation firms.

The occupation-level analysis in Table A5 provides further evidence that the employment effects are concentrated among the highly paid. In high-salary occupations, a one standard deviation increase in LLM exposure is associated with a 5.7 percent reduction in job postings ($p = 0.009$) and £782 salary declines ($p < 0.001$), while low-salary occupations show no significant changes in posting volumes and minimal salary effects.

These results suggest that LLM disruption is almost entirely driven by changes